PIXAR

Scan it. Give it a second life.

An Agentic AI Platform that appraises old electronics, salvages functional motors and sensors for student robotics labs, and certifies zero-landfill circularity.

UN SDG 12: RESPONSIBLE CONSUMPTION | **TARGET 12.5: CIRCULAR SILICON** | **AGENTIC HARDWARE PROTOCOL**

Masterclass Context: What We Built

Designed strictly according to the IBM SkillsBuild Masterclass Framework: Validating the Problem, Assembling the Lean Canvas, and Deploying an Agentic Solution for the UN Sustainable Development Goals.

1. THE TARGET SDG

SDG 12

Responsible Consumption & Production. Specifically Target 12.5: Substantially reduce waste generation through prevention, reduction, recycling and reuse.

2. THE PARADIGM

Agentic AI

Moving beyond static chatbots to autonomous multi-agent pipelines: Visual appraisal, component harvesting planning, local routing, and cryptographic verification.

3. DELIVERABLE

Full App & Deck

Working web application, camera-based AI scanner, 1-page Lean Canvas business model, and verified zero-landfill certification protocol.

The Physical Substrate: India's E-Waste Crisis

Nothing is e-waste until attention is abandoned. When devices stop turning on, valuable micro-components become toxic landfill hazards.

16.0L

TONNES / YEAR

India is the 3rd largest e-waste generator globally, accelerating at 30% CAGR.

78%

UNORGANIZED

Handled by informal scrap yards using toxic open acid-baths and crude open burning.

4.2

DEVICES PER HOME

Average Indian urban household stores 4+ retired phones, tablets, or laptops in cupboards.

94%

SALVAGEABLE

Of discarded phone motors, cameras, microphones, and lithium cells are 100% operational.

Existing Alternatives & Why They Fail

Every current solution treats electronics as either binary working resale or completely dead scrap.

ALTERNATIVE	WHAT THEY DO	CRITICAL LIMITATIONS	PIXAR ADVANTAGE
Local Kabadiwala	Buys scrap electronics for ₹50–₹100 by weight.	Unregulated, burns circuit boards in open acid baths, destroys salvageable motors and camera modules.	Fair valuation, verified certified recycling, scientific harvesting.
Cashify / Trade-In	Appraises and buys working newer smartphones for resale.	Rejects older (>5 yr) or semi-broken phones; zero component recovery; high return rates.	Accepts any condition; appraises down to individual micro-parts.
Passive Drop Bins	Public e-waste collection bins in malls and colleges.	Low participation (<3%), no transparency, no user incentives, untracked logistics.	Instant camera scan valuation, doorstep pickup, cryptographic certificate.

Customer Segments & Early Adopters

Connecting the supply of dormant electronics with immediate, local campus hardware demand.

PRIMARY CUSTOMER SEGMENTS

- **College Students & Households:** Consumers sitting on retired phones and laptops, needing effortless doorstep pickup and eco-conscious disposal.
- **Engineering Maker & Robotics Labs:** Academic labs spending thousands of rupees importing hobby servos, camera modules, and lithium cells.
- **Authorized Recyclers (CPCB Certified):** Formal processing plants struggling to source clean, pre-sorted e-waste feeds from urban centers.
- **Universities & Corporates:** Seeking verifiable EPR (Extended Producer Responsibility) and NAAC sustainability audit compliance.

EARLY ADOPTERS & TIPPING POINT

> **University Hardware Hacker Clubs & Robotics Teams:**
> Engineering students participate in hackathons monthly and have immediate, recurring appetite for low-cost actuators and sensors.

- **IBM SkillsBuild & Lenovo Leap Learners:** Tech-forward students actively training in AI, eager to champion sustainable technology pilots on their campuses.
- **Campus Eco-Clubs:** Organizing student e-waste collection drives with instant scanning kiosks.

Unique Value Proposition

> *"Your Personal Agentic AI Hardware Companion that Appraises, Salvages, Routes, and Certifies Dormant Electronics for a Second Life."*

INSTANT AI APPRAISAL

No complicated checklists. Point phone camera, get instant hardware specs, component health, and fair residual value in under 3 seconds.

CLOSED-LOOP SALVAGE

Vibrator motors become haptic robotics actuators. Smartphone cameras become IoT vision sensors. Zero landfill dumping.

CRYPTOGRAPHIC EPR SEAL

Tamper-proof verifiable QR certificate quantifying avoided carbon (CO_2e) and heavy metal recovery for institutional sustainability audits.

The Multi-Agent AI System

PIXAR coordinates 5 specialized autonomous AI agents across the circular lifecycle.

AGENT 1 • VISION INSPECTOR

Multi-modal Vision Transformer (ViT) & YOLOv8 model detects device model, screen cracks, casing degradation, and port integrity.

AGENT 2 • TEARDOWN PLANNER

Retrieves hardware schematics to generate automated disassembly instructions and isolate high-value reusable subcomponents.

AGENT 3 • MAKER MATCHMAKER

Monitors campus robotics parts requests and routes salvaged motors, displays, and sensors to active student engineering projects.

AGENT 4 • REVERSE LOGISTICS

Coordinates doorstep collection with CPCB-authorized logistics partners, batching pickups for minimum fuel consumption.

AGENT 5 • IMPACT & CERTIFICATE ENGINE

Generates tamper-proof Second-Life Certificates with SHA-256 hashes, computing avoided CO_2e, toxic metals neutralized, and EPR audit compliance tokens.

The End-to-End User Journey

From dormant desk-drawer junk to verified campus second life in five frictionless steps.

01
AI SCAN

Camera scans device; computer vision identifies model and condition with 98% confidence.

02
APPRAISE

System displays hardware health breakdown and estimated residual component value.

03
SALVAGE

Harvestable motors, cameras, and chips are allocated to local campus maker labs.

04
COLLECT

Doorstep courier pickup booked with certified CPCB recycler for remaining housing.

05
CERTIFY

User receives official cryptographic EPR certificate with avoided CO_2e impact metrics.

Acquisition & Distribution Strategy

Leveraging educational networks and frictionless mobile entry points.

ACQUISITION CHANNELS

- **University Maker Clubs:** Direct partnerships with campus robotics societies and IEEE/ACM student branches.
- **IBM SkillsBuild & Lenovo Leap:** Workshops on AI and Circular Economy empowering student ambassadors.
- **Campus E-Waste Drives:** Quarterly college collection events featuring live AI scanning kiosks and leaderboards.
- **Social Media & Campus Influencers:** Video teardown showcases demonstrating cool robots built from salvaged phone parts.

DISTRIBUTION CHANNELS

- **Progressive Web App (PWA):** Instant browser access without app store friction, running client-side camera inference.
- **WhatsApp Bot:** Users snap a photo of old electronics on WhatsApp; AI bot replies with valuation in seconds.
- **Campus Smart Drop Kiosks:** Physical secure drop boxes in engineering department lobbies with integrated digital scale.
- **Hyperlocal Courier Network:** Integration with Dunzo, Porter, or Blinkit APIs for same-day student collection.

Multi-Sided Revenue Streams

Self-sustaining circular economics across B2C, B2B, and institutional buyers.

1. RECYCLER FEES

12–18%

Lead commission from certified CPCB recyclers for aggregated, clean, pre-sorted e-waste feeds.

2. MAKER LAB PASS

₹199/mo

Student maker subscription providing unlimited access to tested salvaged micro-components.

3. B2B EPR AUDITS

Institutional

Annual compliance auditing fees for universities and corporates needing NAAC & CSR sustainability certificates.

4. REFURBISHED PARTS

25–35%

Margin on verified smartphone displays, batteries, and camera modules sold to local repair shops.

Cost Structure & Resource Allocation

Lean operational model optimized for software scalability and campus micro-logistics.

TECHNOLOGY & INFRASTRUCTURE

- Multi-modal Vision API (OpenAI/Gemini/YOLO) inference fees.
- Vector database (Pinecone/Milvus) for component schematics & specs.
- Cloud hosting (Vite, Node.js backend, Cloudflare Edge).
- WhatsApp Business Cloud API & automated notification fees.

OPERATIONS & LOGISTICS

- Subsidized doorstep courier collections for student donors.
- Fabrication and deployment of campus smart drop kiosks.
- Component testing benches and multimeter diagnostic kits for labs.
- Reverse logistics batching to authorized recycler plants.

TEAM & COMPLIANCE

- AI Engineers (Vision fine-tuning & multi-agent orchestration).
- Hardware technicians (Component testing & disassembly SOPs).
- University campus ambassadors & hackathon sponsorship.
- CPCB regulatory audits & legal EPR compliance documentation.

Key Performance Indicators

Tracking both operational venture velocity and tangible environmental impact.

>85%

RECOVERY RATE

Percentage of functional micro-components successfully routed to maker labs rather than crushed.

5,000+

KG LANDFILL DIVERTED

Metric kilograms of heavy-metal electronics diverted from toxic landfills during pilot year.

70%

MAKER COST SAVINGS

Average discount achieved by student inventors using salvaged sensors vs buying new on Amazon.

100%

EPR TRACEABILITY

Every collected device tracked to authorized smelter or maker project via immutable QR hash.

Unfair Advantage: Why PIXAR Wins

Structural moats that cannot be easily copied by traditional scrap dealers or generic trade-in apps.

1. LOCAL CLOSED-LOOP ABSORPTION

Unlike scrap aggregators that must ship waste hundreds of kilometers, PIXAR matches salvaged components directly to makers on the *same university campus*. This eliminates 90% of logistics cost and creates instant localized demand.

2. COMPUTER VISION TEARDOWN DATABASE

Proprietary visual dataset mapping older Indian market phones and laptops to modular salvage potential. The AI knows exactly which motor or camera sensor is inside before the screws are even turned.

3. CRYPTOGRAPHIC EPR AUDIT PROOF

Tamper-proof verifiable QR certificates compliant with India E-Waste (Management) Rules 2022. Provides universities and corporate donors institutional-grade compliance for NAAC and CSR reports.

4. ACADEMIA ECOSYSTEM INTEGRATION

Pre-existing alignment with IBM SkillsBuild, campus robotics labs, and student hacker communities. Trust and adoption are built directly into the student curriculum.

UN SDG 12 Alignment & Future Horizons

Transforming circular consumption from an abstract policy into daily student action.

PHASE 1 • CAMPUS PILOT (M1–M6)

- Launch across 5 engineering campuses.
- Deploy PWA scanner & maker lab deposit kiosks.
- Target: 2,500 devices scanned, 1,200 motors salvaged for student robotics projects.

PHASE 2 • CITY MAKER HUBS (M7–M12)

- Expand to 35 universities & community makerspaces.
- Roll out WhatsApp AI scanning bot.
- Onboard 8 CPCB-certified recycling partners for zero-landfill housing smelting.

PHASE 3 • NATIONAL EPR PROTOCOL (Y2+)

- Partner with major consumer electronics OEMs for automated trade-in & EPR credits.
- Launch national student refurbished hardware marketplace.
- Divert 500+ tonnes of e-waste annually across India.

Join us in building a zero-landfill silicon future.

Explore the live web application: `http://localhost:5173/#/hero` • Contact: `team@pixar-reuse.org`